

Beanstalk BIP-38 Audit Summary

Prepared by Cyfrin

Version 2.3

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Beanstalk Farms on Beanstalk BIP-38. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.3 dated October 13, 2023, was delivered to Beanstalk Farms, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 11 days, the Cyfrin team conducted an audit on the Beanstalk BIP-38 code provided by Beanstalk Farms. The findings consist of 1 High, 1 Medium & 1 Low severity issues with the remainder being informational.

Summary

Project Name	Beanstalk BIP-38
Repository	Beanstalk
Commit	12c608a22535...
Audit Timeline	Sep 12th - Sep 26th, 2023
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	1	0	1
Medium Risk	1	1	0
Low Risk	1	1	0
Informational	13	9	4
Gas Optimizations	0	0	0
Total	16	11	5

6 Audit Scope

Cyfrin conducted an audit of Beanstalk based on the code present in the repository at commit 12c608a, specifically the diff between this commit and c7a20e5, which largely pertains to the BIP-38 migration of the token underlying unripe LP from BEAN:3CRV MetaPool LP to unripe BEAN:ETH Well LP. Other changes introduced in the diff between these two commits, since the previous audit performed by Cyfrin, are also considered in the scope of this audit.

Mitigation remediations have been provided by the Beanstalk Farms team in commit 7606673 contained within the BIP-38 pull request linked above.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Beanstalk is a permissionless algorithmic stablecoin protocol built on Ethereum. The protocol uses a dynamic peg maintenance mechanism to have the price of 1 BEAN (the Beanstalk stablecoin) continuously cross its peg value of 1 USD without collateral requirements.

The BIP-38 upgrade specifically involves migration of the BEAN:3CRV LP Tokens that underlie the Unripe BEAN:3CRV token to BEAN:ETH Well LP Tokens.

The main changes introduced in the diff between the two commits referenced in the audit scope include:

- Replaced Curve integration with Wells integration.
- Added `UnripeFacet::addMigratedUnderlying` to allow for liquidity migration.
- Updated constant values to match on-chain addresses: `C.UNIV3_ETH_USDC_POOL`, `C.UNIV3_ETH_USDT_POOL`, `C.BEANSTALK_PUMP`, `C.BEAN_ETH_WELL`, `DepotFacet::PIPELINE`.
- Changed how Chainlink USDC and USDT Oracles are integrated into the Beanstalk protocol through `LibEthUsdOracle::getEthUsdPrice`.
- Fertilizer is now minted using WETH instead of USDC.
- Added `InitBipBasinIntegration`:
  - Updates earned Stalk per BDV per Season for BEAN and urBEAN3CRV.
  - Whitelists BEAN:ETH Well LP.
- Added `InitMigrateUnripeBean3CrvToBeanEth` which initiates the migration process.
- Modified `MetadataFacet` and `MetadataImage`.
- Pipeline is now able to receive ETH.
- Depot is now able to receive ETH.
- `LibStrings` now supports the string representation of `int256` values less than zero.
- Removed legacy support for withdrawals in `Weather::handleRain`.
- `BeanstalkPrice::price` now considers the price provided by the BEAN:ETH Well and Curve to provide a weighted average price.
- Replaced Curve integration with Well in multiple protocol components, including:
  - `LibFertilizer`
  - `LibUnripeConvert`
  - `BDVFacet`

9 Attestation

Cyfrin attests that it performed a security review of the Beanstalk BIP-38 code provided by Beanstalk Farms.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2023-10-13-cyfrin-beanstalk-bip-38.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Beanstalk Farms has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=0a0312382d8f31c764336fea77b6bdc8848b21e0ecc5434c32503048da70a61b
F=2023-10-13-cyfrin-beanstalk-bip-38-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.